Mohamad Mustafa Abdulkadir

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EDUCATION

United Arab Emirates University Jan. 2021 – May 2025
Bachelor of Science in Computer Science, ***GPA: 3.97 / 4.00*** Al Ain, UAE

SKILLS

Languages: Python, TypeScript, JavaScript, SQL

Frameworks & Libraries: React, Next.js, FastAPI, PyTorch, LangGraph

AI & Machine Learning: LLMs (Training, Fine-tuning, RAG), AI Agents, MCP, LiteLLM, OpenAI/Anthropic APIs, Docling

Cloud, Databases & Tools: AWS, PostgreSQL, Snowflake, Supabase, Qdrant, Docker, Nginx

EXPERIENCE

Mansoura Oct. 2025 – Present
Software Engineer - Data & AI Abu Dhabi, UAE

- Built and launched the company's internal operations platform end to end using React, TypeScript, AWS, PostgreSQL, and Snowflake.
- Developed authentication, role-based access control, backend APIs, document storage and generation, and scheduled cloud workflows.
- Automated operational reporting and added KPI forecasting for management.
- Built an AI agent that matches unidentified rental payments on bank statements to tenants using their payment history, automating a monthly manual process that took the data-entry team hours.

ADIA Lab June 2025 – Sept. 2025
Machine Learning Research Intern Abu Dhabi, UAE

- Trained a transformer-based encoder to predict in-hospital mortality from clinical time-series data derived from EHR records, managing compute-intensive experiments and systematically tracking model performance.
- Built and optimized an end-to-end pipeline that turns longitudinal EHR records into model-ready time series and clinical narratives, cutting projected processing time from 193.4 hours to 18 hours.
- Derived patient-level explanations of the encoder's predictions using attention- and gradient-based attribution, then used them with clinical narratives to fine-tune a transformer decoder for grounded report generation.

Etisalat Jan. 2025 – Apr. 2025
AI Intern Dubai, UAE

- Helped integrate an AI-powered content recommendation engine into the Starz On Business platform.
- Built an AI assistant that resolves tier-one customer support cases from product guides and FAQs.
- Built a dashboard to track the assistant's performance and key operational metrics.

ATRC - STEM Youth Mentorship Program Nov. 2023 – Sept. 2024
Research Assistant Abu Dhabi, UAE

- Researched non-stationary contextual-bandit algorithms and large language models for recommender systems.
- Introduced change-point detectors to contextual bandit algorithms to improve adaptation in stochastic environments.

PROJECTS

GNI

- Agentic platform serving multiple users, where an orchestrator agent delegates tasks to specialist agents and reviews their output before responding.
- Built with Python, FastAPI, LangGraph, and LiteLLM, using Hatchet for durable agent runs and PostgreSQL row-level security to isolate user data.
- Connects agents to external tools through MCP servers and Activepieces, with owner approval before external actions, duplicate-request protection, recovery of interrupted runs after restarts, and per-user usage tracking.

AwesomeRAG

- Full-stack document Q&A workspace with document uploads, OCR, vector retrieval, intent routing, and answers that cite their sources.
- Ran Docling OCR and EmbeddingGemma-300M locally, with Gemini as an optional cloud fallback and Qdrant for vector storage.

German Power Day-Ahead Forecasting Pipeline

- Built an end-to-end pipeline integrating 10 ENTSO-E and EEX datasets, with automated ingestion, normalization, deterministic and LLM-generated quality checks, and hourly feature engineering.
- Benchmarked forecasting models with walk-forward validation, with LightGBM reducing held-out error by 64% versus a same-hour-last-week baseline, and turned hourly forecasts into weekly and monthly baseload, peak, and off-peak fair-value bands.

NutriVision

- Mixed Reality diet-tracking application that identifies food, estimates item weight, and calculates nutritional content.
- Trained deep learning models for weight estimation and fine-tuned YOLOv11 for real-time detection; placed third in the UAEU Innovation Challenge and presented at Expand North Star Dubai 2024.

PUBLICATIONS

- **DoS-based Fake Task Injection for Disrupted Sensing**, AICCSA, 2025.
- **Vault-PMS: A Vault-Based Password Management System**, IWCMC, 2024.

OPEN-SOURCE CONTRIBUTIONS

Deep-ML

- Contributed LeetCode-style problems focused on machine learning and linear algebra.